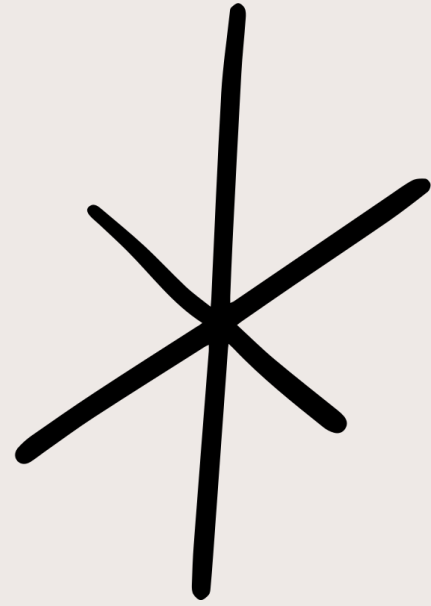


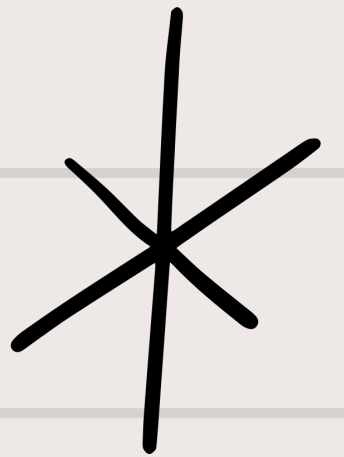
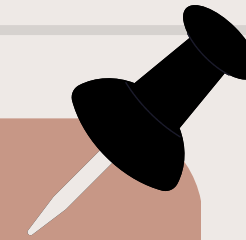
Final Year Project Proposal



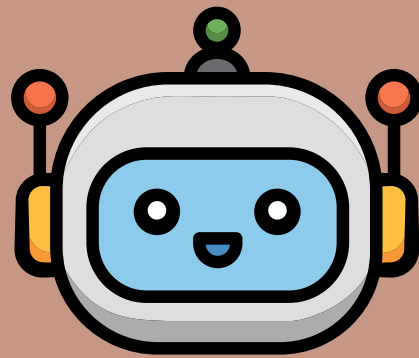
Project Idea Title: 2 | Financial advisor Bot
Module Name: CM3020 Artificial Intelligence



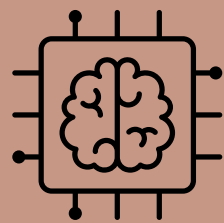
Project Outline



Project aim:



Build and train an AI with financial expertise to advise individual investors. The bot analyses market data using machine learning to produce dynamic investment suggestion, presented via an easily navigable website with plain English explanations so that non-technical users can view, understand and evaluate the advice.



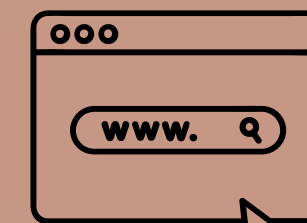
Machine learning

Train models on market data to predict a movement



Market analysis

Analyse real-time and historical stock price data



Web interface

Clean accessible frontend for non-technical user

RobinHood

Black box - gives recommendations with zero explanation. Users cannot evaluate or trust the advice.

Bloomberg Terminal

Costs thousands per year, requires specialist training - completely inaccessible to ordinary investors.

Motivation of this project

Open-source Trading Bots

Built for developers only. No user interface or explanation.
Non - technical users cannot use them.

Human Financial Advisors

Expensive (£150 - 300/hr), not available 24/7, and often influenced by emotions or commission bias.

Main function & Deliverables

1



Automated Data Collection

Fetch real-time & historical stock price data from Yahoo Finance API (OHLCV format)

2



Feature Engineering Pipeline

Calculate technical indicators: Moving Averages, RSI, MACD, Bollinger Bands, Volume Ratio

3



ML Prediction Engine

Random Forest baseline + LSTM neural network - classify each stock as BUY or SELL with confidence score

4



Backtesting System

Evaluate AI strategy on historical data and compare against buy-and-hold baseline.

5



Explanation Generator

Convert model outputs into plain English reasons non-technical users can understand.

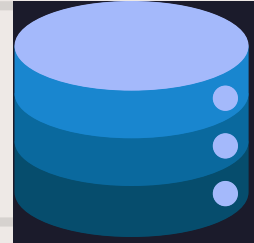
6



Web Interface

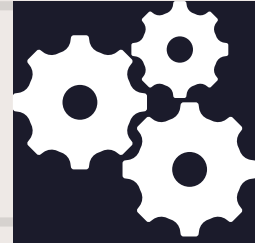
Clean, user-freindly frontend: enter a stock ticker, receive reccomendation + explanation.

Techniques, Tools & Methodology



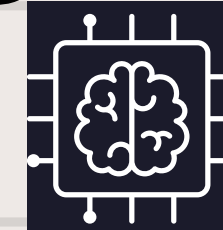
Automated Data Collection

- yfinance - Yahoo finance OHLCV data (python library)
- pandas - data manipulation & preprocessing
- NewsAPI - news sentiment (stretch goal)



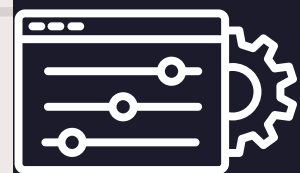
Feature Engineering Pipeline

- Moving Averages (20-day, 50-day)
- RSI, MACD, Bollinger Bands
- Volume Ratio, Price Momentum (1d/5d/20d)



Machine Learning

- scikit-learn: Random Forest (baseline)
- TensorFlow/Keras: LSTM (main focus)
- stable-baselines3: RL (stretch goal)



Backtesting & Evaluation

- Custom backtesting framework
- Total Return, Sharpe Ratio, Max Drawdown
- Time-series CV to prevent data leakage



Explainability

- SHAP values - feature importance
- Plain English explanation generator
- Rule-based indicator interpretation



Web development

- Flask - Python REST API backend
- HTML / CSS / JavaScript - frontend
- Plotly - interactive stock charts

Target Audience



Young First-Time investors

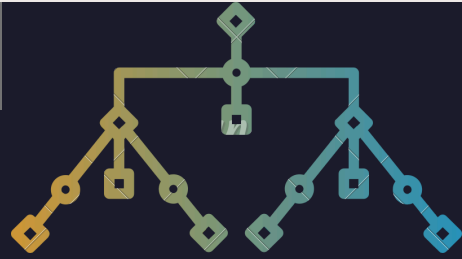
Age 18-30

University students and young professionals who want to start investing but find existing tools too complex or expensive. Need jargon-free advice with clear explanations.

- ✓ Simple BUY/SELL recommendation
- ✓ Plain English explanations
- ✓ Free to use
- ✓ No financial expertise required

Machine Learning Strategy

Phase 1

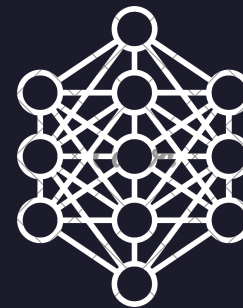


Random Forest

Baseline Model

- ✓ Quick result to validate pipeline
- ✓ Handles tabular feature data well
- ✓ Highly interpretable - good for SHAP
- ✓ Serves as comparison benchmark

Phase 2

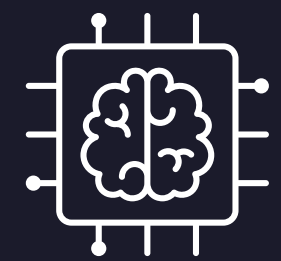


LSTM Network

Main Focus

- ✓ Designed for sequential time-series data
- ✓ captures temporal patterns & trends
- ✓ 60-day look back window
- ✓ Outperforms RF on financial data

Phase 3



Reinforcement Learning

Stretch goal

- ✓ Learns full trading strategy via trial & error
- ✓ Manages portfolio-level decisions
- ✓ Reward = profit, Penalty = loss
- ✓ PPO algorithm via stable-baselines3

Thank You

by Harper Russo

